An Intelligent Aircraft Maintenance Prediction System Using Machine Learning and Real-Time Simulated Data

A synopsis submitted

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Synopsis

1 Introduction

Aircraft maintenance plays a crucial role in ensuring flight safety, operational reliability, and cost-effective airline operations. Conventional maintenance strategies, such as scheduled and corrective maintenance, often result in unnecessary inspections or unexpected component failures, leading to increased operational costs and aircraft downtime. With advancements in Artificial Intelligence (AI) and Machine Learning (ML), predictive maintenance has emerged as a promising approach for forecasting potential failures before they occur, enabling proactive maintenance planning.

The proposed project, "An Intelligent Aircraft Predictive Maintenance System Using Machine Learning and Real-Time Simulated Data," aims to develop an AI-based predictive maintenance framework capable of monitoring aircraft health and predicting the Remaining Useful Life (RUL) of critical components. The system will compare multiple machine learning models to identify the best-performing model for accurate prediction. To demonstrate real-time system functionality, realistic aircraft sensor data will be simulated and continuously processed through an interactive monitoring dashboard. In addition to failure prediction, the system will generate intelligent maintenance recommendations and maintenance priorities to support informed maintenance decision-making.

The significance of this project lies in its ability to integrate predictive analytics with real-time monitoring and decision support into a unified framework. Unlike conventional predictive maintenance systems that focus primarily on failure prediction, the proposed system emphasizes actionable maintenance insights through maintenance recommendations and prioritization. Furthermore, the modular architecture of the proposed framework enables future integration of additional aircraft subsystems, making it a scalable solution for next-generation intelligent aircraft maintenance systems. [1–6].

2 Objectives

Objectives will come here.

1. To collect and preprocess aircraft maintenance or sensor data.
2. To train and evaluate machine learning models for failure prediction.
3. To identify components that require maintenance based on prediction results.
4. To improve prediction accuracy using appropriate feature engineering and model evaluation techniques.
5. To support timely maintenance decisions, reducing maintenance cost and increasing aircraft availability.

3 Literature Review

A literature review was conducted to study recent advancements in aircraft predictive maintenance using Machine Learning and Artificial Intelligence.

Table 1: Literature Review Summary

Sr. No.	**Year**	**Research Paper**	**Summary of Existing Work**	**Research Gap / Proposed Contribution**
1	2025	A Data-Driven Predictive Maintenance Framework for Aero-Engines Using Trans-LSTM and Bayesian Optimization	The paper proposes a Trans-LSTM model with Bayesian Optimization to predict aircraft engine Remaining Useful Life (RUL) using the NASA C-MAPSS dataset, improving maintenance cost and aircraft availability.	Our project focuses on a broader intelligent aircraft predictive maintenance system that can be extended to multiple aircraft components and support real-time maintenance decisions.
2	2025	Predictive Maintenance Programs for Aircraft Engines Based on RUL	Developed a hybrid deep learning model that combines LSTM layers with Transformer self-attention mechanisms to process NASA C-MAPSS engine data. Bayesian Optimization is used to tune hyperparameters and generate optimized maintenance schedules based on Remaining Useful Life (RUL).	The work is limited to aero-engine RUL prediction and fixed maintenance scheduling. Our project aims to develop a scalable aircraft predictive maintenance system capable of monitoring multiple aircraft components while providing intelligent maintenance recommendations.

Sr. No.	Year	Research Paper	Summary of Existing Work	Research Gap / Proposed Contribution
3	2025	Predictive Maintenance Optimization for Industrial Equipment via Reliable Prognosis and Risk-Aware Reinforcement Learning	Built a two-stage decision framework using simulated degradation data. A Probabilistic Neural Network (PNN) predicts Remaining Useful Life (RUL), while a Quantile Regression Deep Q-Network (DQN) reinforcement learning agent determines the most cost-effective maintenance time.	The study focuses on maintenance scheduling for individual industrial components. Our project develops an aircraft-specific predictive maintenance system capable of monitoring multiple aircraft components using real-time simulated sensor data.
4	2024	Aircraft Predictive Maintenance: An Application of Machine Learning Algorithms	Integrated traditional physics-based structural wear equations with Gradient Boosting machine learning algorithms. The model monitors simulated turbine blade telemetry to reduce false alarms and minimize unexpected engine downtime.	The work is limited to turbine blade health prediction using physics-based models. Our project applies machine learning across multiple aircraft components while providing intelligent maintenance prioritization and planning.
5	2023	Predictive Maintenance Analytics and Implementation for Aircraft: Challenges and Opportunities	This paper reviews predictive maintenance techniques, datasets, implementation challenges, and future research opportunities in the aircraft industry, emphasizing AI, automation, and broader industrial adoption.	Our project focuses on implementing a practical machine learning-based predictive maintenance system instead of only reviewing existing techniques. It includes real-time simulated monitoring, Remaining Useful Life prediction, maintenance recommendations, and maintenance prioritization.

3.1 Conventional and Machine Learning Methods

Conventional aircraft maintenance mainly relies on *corrective* and *preventive* maintenance strategies. Corrective maintenance is performed after a component failure, while preventive maintenance is carried out at fixed intervals regardless of the component's actual condition. These methods may increase maintenance costs, cause unnecessary servicing, or result in unexpected failures. Machine Learning overcomes these limitations by analyzing historical maintenance records and aircraft sensor data to predict failures before they occur. Algorithms such as *Decision Tree, Random Forest, Support Vector Machine (SVM), Gradient Boosting,* and *XGBoost* are commonly used to improve maintenance planning, reduce downtime, and enhance aircraft safety and reliability.

3.2 Deep Learning Methods

Deep Learning is an advanced branch of Machine Learning that uses neural networks to learn complex patterns from large datasets automatically. In aircraft predictive maintenance, deep learning models can effectively process sequential sensor data and identify early signs of component degradation. Popular models include *Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Transformer-based models. These methods generally provide higher prediction accuracy and better Remaining Useful Life (RUL) estimation, although they require larger datasets and greater computational resources than traditional machine learning techniques.

4 Methodology

The proposed Aircraft Predictive Maintenance System follows a machine learning-based approach to predict potential aircraft component failures before they occur. The methodology consists of data collection, preprocessing, feature engineering, model training, prediction, and evaluation. The system analyzes aircraft maintenance and sensor data to identify patterns associated with component degradation and provides maintenance recommendations based on prediction results.

4.1 System Architecture

The proposed system consists of the following stages:

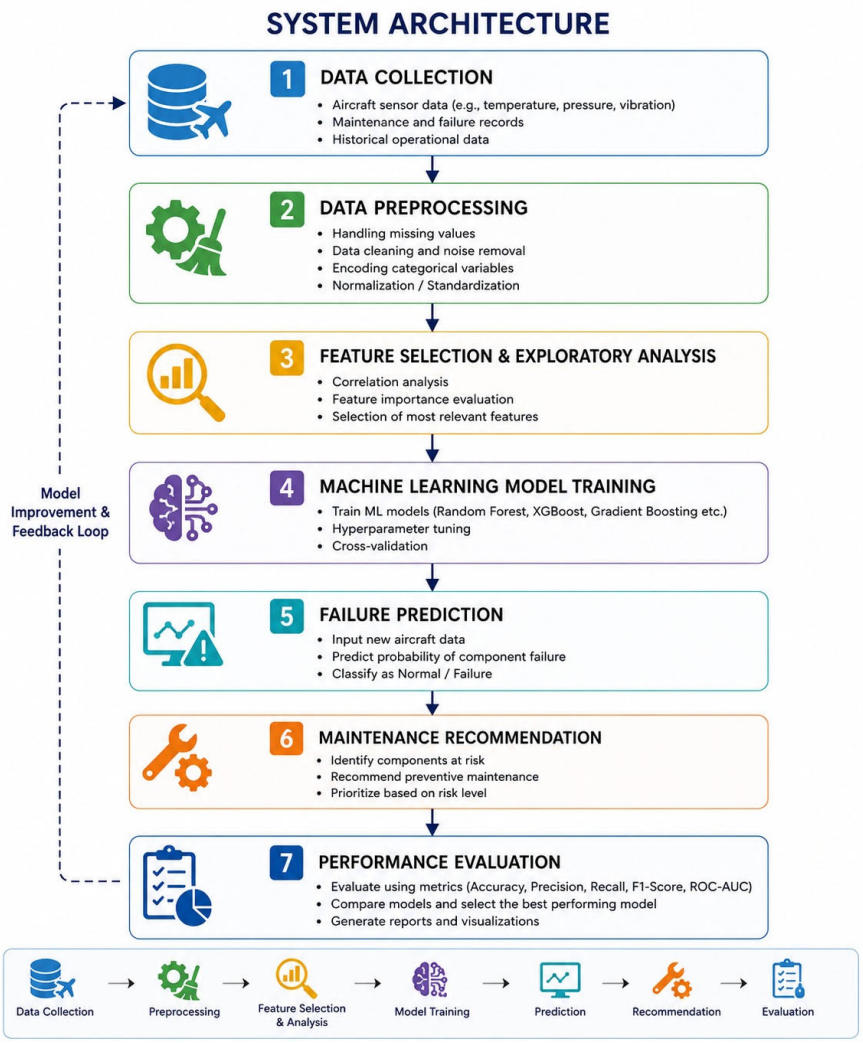


Figure 4.1: System Architecture

4.2 Algorithms

The project will evaluate suitable supervised machine learning algorithms for predictive maintenance. Algorithms such as Random Forest, XGBoost, and Gradient Boosting will be considered for failure prediction due to their high accuracy and ability to handle complex datasets. The best-performing model will be selected based on evaluation metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

4.3 Mathematical Model

The performance of the proposed machine learning model will be evaluated using standard classification metrics based on the confusion matrix.

Let:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

The evaluation metrics are:

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics will be used to compare different machine learning models and determine the best-performing model for aircraft failure prediction.

4.4 Tools, Technologies and Resources

The implementation will use Python as the programming language with machine learning libraries such as Scikit-learn, Pandas, NumPy, and Matplotlib. Model development and experimentation will be carried out using Jupyter Notebook or Google Colab. Git and GitHub will be used for version control, while publicly available aircraft predictive maintenance datasets (such as NASA C-MAPSS or similar datasets) will be used for model training and testing.

4.5 Data Collection and Analysis

Aircraft maintenance and sensor datasets will be collected from publicly available repositories. The collected data will undergo preprocessing steps including handling missing values, normalization, feature selection, and data transformation. Exploratory Data Analysis (EDA) will be performed to identify trends, correlations, and patterns associated with aircraft component failures. The processed data will then be divided into training and testing datasets for model development and performance evaluation.

5 Project Plan and Timeline

The project will be carried out in a phased manner to ensure systematic development and timely completion. Initially, a comprehensive literature review and aircraft maintenance dataset collection will be conducted, followed by data preprocessing and exploratory data analysis (EDA). In the next phase, suitable machine learning models will be developed, trained, and evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Based on the evaluation results, the best-performing model will be selected for aircraft predictive maintenance. The final phase will involve system validation, testing, documentation, and preparation of the final project report and presentation, ensuring the successful completion of the project within the planned academic timeline.

Table 2: Project Timeline

Activity / Months	**Jun.'26**	**Jul.'26**	**Aug.'26**	**Sept.'26**	**Oct.'26**
Problem Statement Identification	✓				
Literature Review	✓				
Requirement Analysis		✓			
Designing		✓			
Experimental Analysis			✓		
Module-wise Implementation			✓	✓	
Testing and Debugging				✓	
Project Report Preparation					✓

6 Expected Outcomes

The proposed project is expected to achieve the following outcomes:

1. Development of an intelligent machine learning-based aircraft predictive maintenance system for failure prediction and Remaining Useful Life (RUL) estimation.
2. Development of a real-time simulated sensor data framework for monitoring the health of critical aircraft components.
3. Identification and deployment of the best-performing machine learning model through comparative performance analysis.
4. Generation of maintenance recommendations and maintenance priorities to support proactive maintenance planning.
5. Development of an interactive web-based dashboard for visualizing aircraft health status, Remaining Useful Life (RUL), prediction results, and maintenance alerts.

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